3000–1800 BCE

The ruins of Uruk, the world's oldest city, are in present-day Iraq. The site of Uruk was first settled around 4800 BCE and became a town around 4000 BCE.

> “CLIMB UPON **THE WALL** OF **URUK,** WALK ALONG IT, I SAY; REGARD THE **FOUNDATION TERRACE** AND EXAMINE THE MASONRY… ”
>
> ***Epic of Gilgamesh,*** Tablet I, *c.*2000 BCE

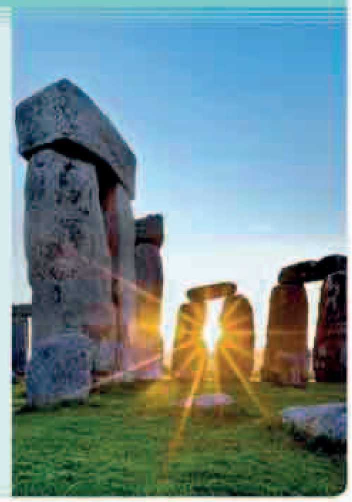
IRRIGATION TECHNIQUES BECAME MORE COMPLEX during the 3rd millennium BCE. The **shadoof was developed** in Mesopotamia in around 2400 BCE. It consisted of an upright frame with a pole suspended from it; on one end of the pole was a bucket for scooping up water, while on the other was a counterweight. By 1350 BCE, the shadoof had spread to Egypt. There, devices called **nilometers had already been developed** to measure the rise and fall of the river, which predicted how good the harvest would be.

In the period 4000–3000 BCE farming communities in Mesopotamia had coalesced to form the **world's first cities**, such as Uruk *c.*3400 BCE. By 3100 BCE, cities had begun to appear in Egypt, beginning with Hierakonpolis. By 2600 BCE, Mohenjo Daro and Harappa, great cities of the Indus Civilization, had been built.

As towns and cities developed, the first **true writing emerged** in Mesopotamia around 3300 BCE, probably prompted by the need to keep detailed records. Originally largely **pictographic**, with signs looking like the things they represented, they were written using a stylus that produced wedge-shaped marks. Cuneiform script developed as the curved outlines of these early signs changed into a series of wedge-shaped lines that gradually became more stylized over time. These symbols were impressed into soft clay, which then hardened to create durable documents. At around the same time, another writing system developed in Egypt. Known as **hieroglyphic**, this system was initially primarily pictographic. The earliest known examples are clay labels from Abydos, *c.*3300 BCE. Writing also developed in the Indus Valley around 2600 BCE, in China by at least 1400 BCE, and in Mesoamerica around 600 BCE.

EARLY ASTRONOMY

Evidence of interest in astronomical phenomena dates from Neolithic times in Europe, when many megaliths were laid out in an orientation that indicated particular lunar or solar events. Some of the stones at Stonehenge (first erected around 2500 BCE) were aligned to indicate the times of year at which the winter and summer solstices occurred. Other features may have been connected with lunar events.



Soda-lime glass was first developed in Mesopotamia around 3500 BCE. It was made by firing silica (sand), soda ash, and lime in a furnace, but initially was suitable only for small objects. In Egypt, **faience became common** from around 3000 BCE. Consisting of a mixture of crushed quartz, calcite lime, and soda lime, which when vitrified produced a blue-turquoise glaze, faience was used by the Egyptians on small sculptures and beads.

The early 3rd millennium BCE saw the **spread of true bronze**, created by alloying copper with tin, which became the most commonly used metal in Mesopotamia between 3000 and 2500 BCE. Clay **crucible furnaces for smelting** appeared there in around 3000 BCE. Mesopotamian metallurgists also invented the technique of **gold granulation** around 2500 BCE. This produced tiny gold balls, which were used to decorate jewelry.

Egyptian boat of the dead
A model of the boat buried near the Great Pyramid of Khufu. The boat was intended to ferry the dead pharaoh's soul across the heavens.

c.3000 BCE Cities begin to appear in Egypt

c.3000 BCE Use of **faience** becomes common in Egypt

c.3000 BCE Clay **crucible furnaces** for smelting appear in Mesopotamia

c.3000 BCE Egyptians develop **sewn-plank ships**

c.2900 BCE Mastaba tombs are built in Egypt

c.2625 BCE Step Pyramid of Djoser is built in Egypt

c.2600 BCE Cities of Mohenjo Daro and Harappa are established by the Indus Civilization

2550–2472 BCE Great Pyramids of Giza are built in Egypt